

Fixed Point Theorems in Partially Ordered Sets

Key Concepts:

1. Partially Ordered Set $(X, \leq)$:
 - Reflexive: $x \leq x$
 - Transitive: $x \leq y, y \leq z \Rightarrow x \leq z$
 - Anti-symmetric: $x \leq y, y \leq x \Rightarrow x = y$
2. Monotone (Isotone) Mapping $f: X \rightarrow X$
 - Nondecreasing: $x \leq y \Rightarrow f(x) \leq f(y)$
 - Nonincreasing: $x \leq y \Rightarrow f(x) \geq f(y)$
3. Weakly Contractive Mapping:
 $||Tx - Ty|| \leq ||x - y|| - \psi(||x - y||)$
where $\psi(0) = 0$, ψ continuous, nondecreasing
4. Convergence Theorem:
If $x_0 \leq f(x_0)$, then $x_n = f(x_{n-1})$ is nondecreasing and converges to fixed point x^*
5. Uniqueness Conditions (SC1)-(SC3):
 - (SC1) Every pair has a lower bound or upper bound
 - (SC2) Sequences have comparable subsequences
 - (SC3) f maps comparable elements to comparable elements